

Pakhi Banchalia

Dresden, Germany | pakhibanchalia24@gmail.com | [GitHub](#) | [Portfolio](#)

EDUCATION

- **Technische Universität Dresden** Oct 2025 - Present
Master of Science in Computational Modeling and Simulation (Applied AI track) Dresden, Germany
- **Amrita Vishwa Vidyapeetham Kerala, India** 2021-2025
Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) Amritapuri, India

RESEARCH EXPERIENCE

- **Aalto University** Jun 2026 - Aug 2026
Summer Research Intern Espoo, Finland
 - One of 67 interns selected from 5,500 applications across 133 countries for the AScI International Summer Research Programme.
 - Surveyed empowerment-based assistance in multi-agent reinforcement learning for open problems.
 - Found that AvE (Assistance via Empowerment, NeurIPS 2020) tests assistance with an agent acting remotely rather than embodied; rebuilt the experiment with an embodied assistant to measure what that costs.
- **Lossfunk** Jul 2025 - Sep 2025
Research Intern Bangalore, India
 - Built and evaluated homeostatic reinforcement-learning agents in the Crafter environment, using multiple internal variables such as hunger, energy, and safety to shape agent behaviour.
 - Tested how alternative reward formulations and competing drives affect exploration, survival, behavioural diversity, and policy stability, against standard RL baselines.
- **Computational Machinery of Cognition (CMC) Lab, TU Dresden** Nov 2024 - Present [Intermittent]
Research Intern (Independent Collaboration) Dresden, Germany
 - Designed RL agents with internal physiological drives to study retention and generalization across changing task distributions.
 - Explored foraging behaviour in deep RL agents, to study and understand any state-dependent dynamic coupling between their internal state and motor decisions; signatures absent under conventional RL
 - For a separate project, implemented a reward-modulated, gradient-free learning algorithm based on weight perturbations to investigate biologically plausible credit assignment without backpropagation.
- **Google Summer of Code, INCF** May 2024 - Sept 2024
Contributor Remote
 - Simulated *C. elegans* neural development using Neural Cellular Automata and evolutionary optimization, representing developmental changes in cell state and connectivity over time.
 - Modeled evolving neural connectivity with Graph Neural Networks and performed Topological Data Analysis with Kepler Mapper to inspect developmental and connectivity trajectories.
 - Delivered reproducible, documented open-source research code for computational developmental neuroscience. [\[Code\]](#)

LEADERSHIP

- **amFOSS** Sep 2022 - Jul 2025
Core Committee Member; Head of Design Amritapuri, India
 - On a five-person core committee for 50-55 active members: ran weekly project check-ins, surfaced stalled work, and bridged the founder, university staff, and members.
 - Led the design team, reviewing work and making final design decisions, including the amfoss.in redesign.
 - Mentored junior cohorts and ran performance reviews tied to continued membership.

PUBLICATIONS

- Pillai, H.S., **Banchalia, P.**, Jamboji, V., Krishnan, S.R. (2026). *CAN: Continuously Adapting Network*. In *Data Science and Applications, ICDSA 2025. Lecture Notes in Networks and Systems*, vol 1723. Springer, Cham. [\[DOI\]](#)

PROJECTS

- **Alignd: Sponsorship Agent for TinyFish Accelerator**

- One of 60 demos selected from over 2,000 applications to the TinyFish Accelerator, a \$2 million program run by TinyFish and Mango Capital. Reached Phase 2.
- Built an AI agent that navigates live web sources to find and evaluate sponsorship opportunities, and designed the relevance and outreach workflow around it. [\[Code\]](#)

- **SEANN: Self-Expanding and Adaptive Neural Networks**

- Extended the CAN continual-learning framework, which mitigates catastrophic forgetting by preserving significant task-specific connections through selective neuron freezing and Hebbian learning principles inspired by synaptic plasticity.
- Introduced an alternative to conventional model scaling that dynamically extends network capacity and identifies task-relevant subnetworks, reducing interference between tasks and catastrophic forgetting.

TALKS AND POSTERS

- **Brain-like features in homeostatic deep reinforcement learning agents @ Bernstein Conference 2025**

- Poster presentation on homeostatic RL agents integrating internal and sensory inputs, showing structured behavioral and neural patterns

- **Emergence of Brain-like Features in Homeostatic RL Agents @ BrainBody 2025**

- Oral presentation; demonstrated coupling between internal states, neural dynamics, and adaptive exploration

TECHNICAL SKILLS

- **Languages:** Python, C++, C, Java, CSS, HTML, JavaScript, SQL, MATLAB
- **ML / Research:** PyTorch, TorchVision, Gymnasium (OpenAI Gym), JAX (familiar)
- **Scientific Computing:** NumPy, SciPy, Pandas, Matplotlib, Kepler Mapper (TDA)
- **Development:** Git, Linux/Unix, Docker, Weights & Biases (W&B), Docker, Django
- **Research Areas:** Agentic AI, Reinforcement Learning, Biologically-Inspired Learning, Robotics, Neural Cellular Automata, Continual Learning, GNNs